CH 8 — Taylor Polynomial and Taylor Theorem

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Recall: For a single variable function, $f(x)$, the second-order Taylor Polynomial at $x = a$ is

$$P_{2,a}(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 = L_a(x) + \frac{1}{2}f''(a)(x-a)^2$$

This is the 2nd order polynomial which agrees with $f(x)$ at $x = a$ up to the 2nd order derivative, i.e.

$$P_{2,a}(a) = f(a)$$
$$P'_{2,a} = f'(a)$$
$$P''(2,a)(a) = f''(a)$$

Definition — Taylor Polynomial

If $f(x,y) \in C^2$ at (a,b), then second-order Taylor Polynomial is defined of f at (a,b) by

$$P_{2,(a,b)}(a,b) = L_{(a,b)}(x,y) + \frac{1}{2}\big(f_{xx}(x-a)^2 + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(y-b)^2\big)$$

Definition — Hessian Matrix

The **Hessian Matrix** of $f(x,y)$ is

$$Hf(x,y) = \begin{bmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{yx}(x,y) & f_{yy}(x,y) \end{bmatrix}$$

Examples:

1. Use both linearization and the second order Taylor Polynomial of $f(x,y) = e^{x+y}$ at $(0,0)$ to approximate $f(0.1, 0.2)$

   $L_{(0,0)}(0.1,0.2) = 1 + x + y\Big|_{(x,y)=(0.1,0.2)} = 1.3$

   $P_{2,(0,0)} = L_{(0,0)}(0.1,0.2) + \frac{1}{2}((x)^2 + 2(x)(y) + (y)^2)\Big|_{(x,y)=(0.1,0.2)}$

   $= 1.3 + \frac{1}{2}((0.1)^2 + 2(0.1)(0.2) + (0.2)^2) = 1.3 + (0.005 + 0.02 + 0.02) = 1.345$

2. Use the linearizatio and second-order Taylor polynomial of $f(x,y) = \frac{1}{2}x - y + x^5$ at $(0,0)$ to approximate $f(0.1, 0.1)$

   $L_{0,0}(x,y) = 0 + \big(\frac{1}{2} + 5(0)^4\big)(x) + (-1)(y) = \frac{1}{2}x - y$

   $L_{0,0}(0.1,0.1) = \frac{1}{2}(0.1) - 0.1 = -0.05$

   $P_{2,(0,0)}(x,y) = -0.05 + \frac{1}{2}[(20(0)^3)(x)^2 + 2(0)(x)(y) + (0)(y)^2] = -0.05$

   $P_{2,(0,0)} = -0.05$

In practical application, we don't know the value of what we are trying to approximate, so we cannot calculate the approximate the approximation error exactly.

Theorem — Taylor's Polynomial 1D

If $f''(x)$ exists on $[a, x]$, then $\exists a \in \mathbb{R}$ s.t $c \in (a, x)$ and

$$f(x) = f(a) + f'(a)(x-a) + R_{1,a}(x)$$

where $R_{1,a}(x) = \frac{1}{2}f''(c)(x-a)^2$

Theorem — Taylor's Polynomial 2D

If $f(x, y) \in C^2$ in some neighbourhood $N(a, b)$ of (a, b), then $\forall (a, b) \in N(a, b), \exists (c, d)$ on the line segment joining (a, b) and (x, y) s.t.

$$f(x, y) = f(a, b) + f_x(a, b)(x-a) + f)y(a, b)(y-b) + R_{1,(a,b)}(x, y)$$

where

$$R_{1,(a,b)}(x, y) = \frac{1}{2}\left[f_{xx}(c, d)(x-a)^2 + 2f_{xy}(c, d)(x-a)(y-b) + f_{yy}(c, d)(y-b)^2\right]$$

Theorem — Corollary of Upper Bound of 2D Tayloyr Polynomial

If $f(x, y) \in C^2$ in some neighbourhood $N(a, b)$ of (a, b), then $\exists M \in \mathbb{R}_{\geq 0}$ s.t.

$$R_{2,(a,b)}(x, y) \leq M\|(x, y) - (a, b)\|^2 \quad \forall (x, y) \in N(a, b)$$